Chapter 8 — nth Roots & Rational Exponents

Algebra 1 • Complete Study Guide: Concept to Practice

What You Will Learn: (1) Where radical notation comes from. (2) Find nth roots and understand the four cases. (3) Evaluate and simplify rational exponents. (4) Apply exponent properties. (5) Solve real-life problems.

Section 1 — Where Does the Radical Sign Come From?

Key Insight: Radicals Are Like Long Division

The radical sign $\sqrt{\ }$ grew out of the idea of **long division**! Just as long division *undoes* multiplication, taking a root *undoes* exponentiation.

Step 1 — Start with multiplication:

$$6 = 2 \times 3$$

Step 2 — Division undoes it:

$$2\overline{)\,6} \quad \Rightarrow \quad 3$$

Step 3 — Scale up to exponentiation:

$$\underbrace{a \cdot a \cdot a \cdots a}_{10 \text{ times}} = 1024 \qquad \Longrightarrow \qquad \sqrt[10]{1024} = a = 2$$

$$2\overline{)\,1024} \qquad \longrightarrow \qquad \sqrt[10]{1024} = 2$$

Operation	Symbol	Meaning
Multiplication	$2 \times 3 = 6$	Combine equal groups
Long Division	$2\overline{)\,6}$	**Undo** multiplication
Exponentiation	$2^{10} = 1024$	Repeated multiplication
Radical	$\sqrt[10]{1024} = 2$	**Undo** exponentiation

The Big Picture

Three operations — same relationship, different unknown:

$$\underbrace{b^p = x}_{\text{exponentiation: find } x} \iff \underbrace{b = \sqrt[p]{x}}_{\text{radical: find } b} \iff \underbrace{p = \log_b x}_{\text{logarithm: find } p}$$

Exponentiation, radicals, and logarithms are the same triangle of ideas.

Section 2 — Core Definition of Rational Exponents

Core Concept: The Key Formula

$$\boxed{a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m}$$

Denominator of the exponent = the **Root index**

Numerator of the exponent = the **Power**

Special case ($m = 1$): $a^{1/n} = \sqrt[n]{a}$

Connection to rational exponents via the Power of a Power property:

$$(a^{1/2})^2 = a^{(1/2)\cdot 2} = a^1 = a, \quad (a^{1/3})^3 = a^1 = a, \quad (a^{1/4})^4 = a^1 = a$$

Therefore $a^{1/n}$ *is* the nth root of a.

Section 3 — Finding nth Roots — The Four Cases

Core Concept: Real nth Roots of a

Let $n > 1$ be an integer and a a real number.

Condition	Real roots	Count	Example
n **odd**, any a	$\sqrt[n]{a}$	One	$\sqrt[3]{-8} = -2$
n **even**, $a > 0$	$\pm\sqrt[n]{a}$	Two	$\pm\sqrt[4]{16} = \pm 2$
n **even**, $a = 0$	0	One	$\sqrt[4]{0} = 0$
n **even**, $a < 0$	(none)	Zero	$\sqrt[4]{-16}$ undefined

$\pm\sqrt[n]{a}$ represents *both* the positive and negative nth roots. An even index with a negative radicand produces **no real number**.

Worked Example: Identifying and Finding nth Roots

Find each root, or state if not a real number.

(a) $\sqrt[3]{27}$ **(b)** $\sqrt[3]{-64}$ **(c)** $\pm\sqrt[4]{625}$ **(d)** $\sqrt[4]{-81}$ **(e)** $\sqrt[5]{-243}$

(a) $\sqrt[3]{27} = \sqrt[3]{3^3} = 3$ — *odd index, positive radicand*

(b) $\sqrt[3]{-64} = \sqrt[3]{(-4)^3} = -4$ — *odd index allows negative radicand*

(c) $\pm\sqrt[4]{625} = \pm\sqrt[4]{5^4} = \pm 5$ — *even index, $a > 0$: two roots*

(d) $\sqrt[4]{-81}$ is **not a real number** — *even index, $a < 0$*

(e) $\sqrt[5]{-243} = \sqrt[5]{(-3)^5} = -3$ — *odd index: one real root*

Section 4 — Evaluating Expressions of the Form $a^{1/n}$

Core Concept: Key Distinction: Where Is the Negative Sign?

Expression	Meaning	Value	
$8^{1/3}$	$\sqrt[3]{8}$	2	
$(-8)^{1/3}$	$\sqrt[3]{-8}$	-2	(odd index: OK)
$-8^{1/3}$	$-(\sqrt[3]{8})$	-2	(negate after rooting)
$(-16)^{1/4}$	$\sqrt[4]{-16}$	*not real*	(even index, $a < 0$)
$-16^{1/4}$	$-(\sqrt[4]{16})$	-2	(negate after rooting)

Worked Example: Evaluating $a^{1/n}$ — Sign Placement

(a) $(-8)^{1/3}$ **(b)** $-\sqrt[3]{8}$ **(c)** $16^{1/4}$ **(d)** $(-16)^{1/4}$

(a) $(-8)^{1/3} = \sqrt[3]{(-2)^3} = -2$ — *odd index; root of a negative number*

(b) $-\sqrt[3]{8} = -(\sqrt[3]{2^3}) = -2$ — *negate the principal root*

(c) $16^{1/4} = \sqrt[4]{2^4} = 2$ *positive principal fourth root*

(d) $(-16)^{1/4}$ is **not a real number** *even index with negative base*

Section 5 — Evaluating $a^{m/n}$ — Two Methods

Core Concept: Root First vs. Power First

$a^{m/n} = \left(\sqrt[n]{a}\right)^m$ **(Method 1: root first)** $=$ $\sqrt[n]{a^m}$ **(Method 2: power first)**

Expression	Method 1: Root First	Method 2: Power First
$8^{2/3}$	$(\sqrt[3]{8})^2 = 2^2 = 4$	$\sqrt[3]{8^2} = \sqrt[3]{64} = 4$
$16^{3/4}$	$(\sqrt[4]{16})^3 = 2^3 = 8$	$\sqrt[4]{16^3} = \sqrt[4]{4096} = 8$

Tip: Root first is almost always easier — it keeps numbers small before you raise to a power.

Worked Example: Full Set of Worked Examples

(1) $9^{1/2}$ **(2)** $27^{2/3}$ **(3)** $32^{3/5}$ **(4)** $(-8)^{2/3}$ **(5)** $4^{-3/2}$ **(6)** $\left(\frac{4}{9}\right)^{3/2}$ **(7)** $(-27)^{2/3}$ **(8)** $81^{-3/4}$

(1) $9^{1/2} = \sqrt{9} = \boxed{3}$ *square root*

(2) $27^{2/3} = (\sqrt[3]{27})^2 = 3^2 = \boxed{9}$ *cube root first, then square*

(3) $32^{3/5} = (\sqrt[5]{32})^3 = 2^3 = \boxed{8}$ *fifth root first, then cube*

(4) $(-8)^{2/3} = (\sqrt[3]{-8})^2 = (-2)^2 = \boxed{4}$ *odd index allows negative base*

(5) $4^{-3/2} = \dfrac{1}{4^{3/2}} = \dfrac{1}{(\sqrt{4})^3} = \dfrac{1}{2^3} = \boxed{\dfrac{1}{8}}$ *negative exponent ⇒ reciprocal*

(6) $\left(\frac{4}{9}\right)^{3/2} = \left(\sqrt{\frac{4}{9}}\right)^3 = \left(\frac{2}{3}\right)^3 = \boxed{\frac{8}{27}}$ *root the fraction, then cube*

(7) $(-27)^{2/3} = (\sqrt[3]{-27})^2 = (-3)^2 = \boxed{9}$ *odd index; squaring makes positive*

(8) $81^{-3/4} = \dfrac{1}{81^{3/4}} = \dfrac{1}{(\sqrt[4]{81})^3} = \dfrac{1}{3^3} = \boxed{\dfrac{1}{27}}$ *reciprocal, then root first*

Section 6 — Negative Rational Exponents

Core Concept: Negative Exponent Rule

$$\boxed{a^{-m/n} = \frac{1}{a^{m/n}}}, \qquad a \neq 0$$

Strategy: flip to reciprocal first, then evaluate $a^{m/n}$ as usual.

Example: $27^{-2/3} = \dfrac{1}{27^{2/3}} = \dfrac{1}{(\sqrt[3]{27})^2} = \dfrac{1}{3^2} = \dfrac{1}{9}$

Section 7 — Exponent Properties with Rational Exponents

Core Concept: All Five Laws Still Apply

Property	Rule	Example
Product of powers	$a^m \cdot a^n = a^{m+n}$	$8^{1/3} \cdot 8^{2/3} = 8^1 = 8$
Quotient of powers	$\dfrac{a^m}{a^n} = a^{m-n}$	$\dfrac{16^{3/4}}{16^{1/4}} = 16^{1/2} = 4$
Power of a power	$(a^m)^n = a^{mn}$	$(9^{1/2})^4 = 9^2 = 81$
Power of a product	$(ab)^m = a^m b^m$	$(4 \cdot 9)^{1/2} = 2 \cdot 3 = 6$
Power of a quotient	$\left(\dfrac{a}{b}\right)^m = \dfrac{a^m}{b^m}$	$\left(\dfrac{8}{27}\right)^{1/3} = \dfrac{2}{3}$

Worked Example: Simplifying with Exponent Properties

(a) $5^{1/3} \cdot 5^{2/3}$ **(b)** $\dfrac{x^{3/4}}{x^{1/4}}$ **(c)** $(x^{2/3})^{3/2}$ **(d)** $\left(\dfrac{27}{8}\right)^{2/3}$

(a) $5^{1/3} \cdot 5^{2/3} = 5^{1/3+2/3} = 5^1 = 5$ *product of powers*

(b) $\dfrac{x^{3/4}}{x^{1/4}} = x^{3/4-1/4} = x^{1/2}$ *quotient of powers*

(c) $(x^{2/3})^{3/2} = x^{(2/3)(3/2)} = x^1 = x$ *power of a power*

(d) $\left(\dfrac{27}{8}\right)^{2/3} = \dfrac{27^{2/3}}{8^{2/3}} = \dfrac{3^2}{2^2} = \dfrac{9}{4}$ *power of a quotient, then root first*

Section 8 — Converting Between Radical and Exponential Forms

Core Concept: Conversion at a Glance

$$\sqrt[n]{a^m} = a^{m/n} \qquad \longleftrightarrow \qquad a^{m/n} = \sqrt[n]{a^m}$$

Radical form	Exponential form
$\sqrt[3]{x^2}$	$x^{2/3}$
$(\sqrt[4]{z})^3$	$z^{3/4}$
$\dfrac{1}{\sqrt[3]{a^2}}$	$a^{-2/3}$
$\sqrt[3]{x^6}$	$x^{6/3} = x^2$

Worked Example: Converting Forms

(a) Write $\sqrt[4]{x^3}$ in exponential form.
(b) Write $a^{5/6}$ in radical form.
(c) Write $\dfrac{1}{\sqrt[3]{y^4}}$ with a rational exponent.
(d) Simplify $\sqrt[3]{x^6}$.

(a) $\sqrt[4]{x^3} = x^{3/4}$ *index → denominator, power → numerator*

(b) $a^{5/6} = \sqrt[6]{a^5}$ *denominator is index, numerator is power*

(c) $\dfrac{1}{\sqrt[3]{y^4}} = \dfrac{1}{y^{4/3}} = y^{-4/3}$ *reciprocal gives negative exponent*

(d) $\sqrt[3]{x^6} = x^{6/3} = x^2$ *convert, simplify the fraction*

Section 9 — Practice Problems

Level 1 — Evaluate (nth Roots and $a^{1/n}$)

1. $\sqrt[3]{125}$

2. $\sqrt[3]{-216}$

3. $\pm\sqrt[4]{256}$

4. $\sqrt[6]{-64}$

5. $64^{1/3}$ (write as a radical first)

6. $25^{3/2}$

7. $81^{3/4}$

8. $(-32)^{1/5}$

Level 2 — Negative Exponents and Mixed Forms

1. $16^{-1/4}$

2. $8^{-2/3}$

3. $27^{-4/3}$

4. $81^{-3/4}$

5. $(-32)^{3/5}$

6. $\left(\frac{4}{9}\right)^{3/2}$

7. $\left(\frac{8}{27}\right)^{2/3}$

8. $(-8)^{4/3}$

Level 3 — Exponent Properties and Conversion

1. $8^{1/3} \cdot 8^{2/3}$

2. $\dfrac{16^{3/4}}{16^{1/4}}$

3. $4^{1/2} \cdot 9^{1/2}$

4. $(9^{1/2})^4$

5. $7^{1/4} \cdot 7^{3/4}$

6. $\dfrac{x^{5/6}}{x^{1/6}}$ (simplify)

7. $(y^{3/4})^{8/3}$ (simplify)

8. $\left(\dfrac{64}{125}\right)^{2/3}$

Level 4 — Converting Forms and Solving Equations

1. Write $\sqrt[5]{m^3}$ in exponential form.

2. Write $(\sqrt[4]{p})^7$ in exponential form.

3. Write $\dfrac{1}{\sqrt[6]{k^5}}$ using a rational exponent.

4. Evaluate $125^{2/3}$ by rewriting in radical form first.

5. Evaluate $243^{3/5}$ by rewriting in radical form first.

6. Solve for x: $x^{3/4} = 27$

7. Solve for x: $x^{2/3} = 16$

8. Solve for x: $x^{-1/2} = \dfrac{1}{3}$

Section 10 — Real-Life Applications

Core Concept: Strategy

1. Write the formula and identify the variable with a rational exponent.
2. Raise both sides to the **reciprocal** exponent to isolate the variable.
3. Simplify and check that the answer makes sense in context.

Worked Example: Real-Life Problems

(a) *Geometry:* The edge of a cube with volume V is $s = V^{1/3}$. Find s when $V = 343$ cm^3.

(b) *Physics:* Pendulum period $T = 2\pi\left(\dfrac{L}{9.8}\right)^{1/2}$ s. Find T when $L = 2.45$ m.

(a) $s = 343^{1/3} = \sqrt[3]{7^3} = 7$ cm *cube root*

(b)

$$T = 2\pi\left(\tfrac{2.45}{9.8}\right)^{1/2} = 2\pi(0.25)^{1/2} = 2\pi(0.5) = \pi \approx 3.1 \text{ s}$$

Level 5 — Real-Life Problems

1. A cube has volume 512 in^3. Find its edge length using $s = V^{1/3}$.

2. The radius of a sphere: $r = \left(\dfrac{3V}{4\pi}\right)^{1/3}$. Find r when $V = \dfrac{4\pi}{3}$ cubic units.

3. Skid-mark speed: $v = (24d)^{1/2}$ mph. Find v when $d = 6$ ft.

4. Solve: $x^{3/4} = 27$.

5. Solve: $x^{2/3} = 16$.

Complete Solutions — Chapter 8

Level 1 — nth Roots and $a^{1/n}$

1. $\sqrt[3]{125} = 5$ **2.** $\sqrt[3]{-216} = -6$ **3.** $\pm\sqrt[4]{256} = \pm 4$ **4.** Not a real number (even index, $a < 0$)

5. $64^{1/3} = \sqrt[3]{64} = 4$ **6.** $25^{3/2} = (\sqrt{25})^3 = 5^3 = 125$ **7.** $81^{3/4} = (\sqrt[4]{81})^3 = 3^3 = 27$
8. $(-32)^{1/5} = -2$

Level 2 — Negative Exponents and Mixed Forms

1. $16^{-1/4} = \dfrac{1}{\sqrt[4]{16}} = \dfrac{1}{2}$ **2.** $8^{-2/3} = \dfrac{1}{(\sqrt[3]{8})^2} = \dfrac{1}{4}$ **3.** $27^{-4/3} = \dfrac{1}{(\sqrt[3]{27})^4} = \dfrac{1}{81}$

4. $81^{-3/4} = \dfrac{1}{(\sqrt[4]{81})^3} = \dfrac{1}{27}$ **5.** $(-32)^{3/5} = (\sqrt[5]{-32})^3 = (-2)^3 = -8$

6. $\left(\dfrac{4}{9}\right)^{3/2} = \left(\dfrac{2}{3}\right)^3 = \dfrac{8}{27}$ **7.** $\left(\dfrac{8}{27}\right)^{2/3} = \dfrac{(\sqrt[3]{8})^2}{(\sqrt[3]{27})^2} = \dfrac{4}{9}$ **8.** $(-8)^{4/3} = (\sqrt[3]{-8})^4 = (-2)^4 = 16$

Level 3 — Exponent Properties

1. $8^{1/3} \cdot 8^{2/3} = 8^1 = 8$ **2.** $\dfrac{16^{3/4}}{16^{1/4}} = 16^{1/2} = 4$ **3.** $4^{1/2} \cdot 9^{1/2} = (36)^{1/2} = 6$ **4.** $(9^{1/2})^4 = 9^2 = 81$

5. $7^{1/4} \cdot 7^{3/4} = 7^1 = 7$ **6.** $\dfrac{x^{5/6}}{x^{1/6}} = x^{4/6} = x^{2/3}$ **7.** $(y^{3/4})^{8/3} = y^2$ **8.** $\left(\dfrac{64}{125}\right)^{2/3} = \dfrac{16}{25}$

Level 4 — Converting and Solving

1. $m^{3/5}$ **2.** $p^{7/4}$ **3.** $k^{-5/6}$

4. $125^{2/3} = (\sqrt[3]{125})^2 = 5^2 = 25$ **5.** $243^{3/5} = (\sqrt[5]{243})^3 = 3^3 = 27$

6. $x^{3/4} = 27 \Rightarrow x = 27^{4/3} = (\sqrt[3]{27})^4 = 3^4 = 81$
7. $x^{2/3} = 16 \Rightarrow x = 16^{3/2} = (\sqrt{16})^3 = 4^3 = 64$
8. $x^{-1/2} = \dfrac{1}{3} \Rightarrow x^{1/2} = 3 \Rightarrow x = 9$

Level 5 — Real-Life Problems

1. $s = 512^{1/3} = 8$ inches

2. $r = \left(\dfrac{3 \cdot \frac{4\pi}{3}}{4\pi}\right)^{1/3} = (1)^{1/3} = 1$ unit

3. $v = (24 \times 6)^{1/2} = 144^{1/2} = 12$ mph

4. $x^{3/4} = 27 \Rightarrow x = 81$

5. $x^{2/3} = 16 \Rightarrow x = 64$